

Akshay Trikha

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EXPERIENCE

QuantumScape

Machine Learning Engineer, Office of the CTO

09/21 – Present

San Francisco, CA

- Building and buying LLM-related software. Led company-wide effort to create an “AI automation wishlist”.
- Trained and deployed PyTorch vision models for manufacturing defect detection, serving 1M inferences/month.
- Created a segmentation model that runs on >60 camera systems at the company (tinyurl.com/finding-mask).

Sandia National Laboratories

Researcher, 9-person team

09/20 – 05/21

San Francisco, CA

- Investigated link between diameter of ferroelectric barium titanate nanoparticles and dielectric constant.
- Developed a Python image-processing pipeline using OpenCV, NumPy, and Matplotlib to extract particle sizes and distributions from TEM images, optimized runtime by 25x using Numba.
- Presented at Materials Research Society '21 Spring Meeting & published in MRS Advances (tinyurl.com/sandia-paper).

AMISTAD Lab

Researcher, 6-person team

05/19 – 12/19

Claremont, CA

- Explored why machine learning works from an information theory and search perspective.
- Co-authored *The Bias-Expressivity Tradeoff*, won best paper award at ICAART2020 in Valletta, Malta.
- Co-authored *The Futility of Bias-Free Learning*, presented at AI2019 in Adelaide, Australia (tinyurl.com/amistad-futility).

PROJECTS

Scaling Laws for Interatomic Potentials | PyTorch

Berkeley, CA

03/24 – 05/25

- Investigated scaling laws comparing transformers and EquiformerV2 for learning physics from data.
- Trained with 8 GPUs up to 10M parameter models on 1M datapoints from Open Materials 2024 dataset.
- Wrote about findings here (tinyurl.com/materials-scaling).

Neural Style Transfer | JavaScript, React, HTML/CSS

San Francisco, CA

07/21

- Created a neural style transfer web app generating stylized images from webcam input in real-time using a pretrained TensorFlow.js model (styletransfer.art).

GPT-2 Trump | Reimplemented with Hugging Face Transformers

San Francisco, CA

12/20

- Finetuned GPT-2 774M using ~56,500 Trump tweets to understand model's capability for persuasion. (tinyurl.com/gpt2-trump)

SKILLS

Technical: Python (PyTorch, TensorFlow, NumPy, SciPy, Scikit-learn, Pandas, OpenCV), C++, C, JavaScript (TensorFlow.js), React, Vue, SQL, HTML/CSS, Java

Natural Language: Hindi (fluent), Mandarin (conversational), Sanskrit (learning), English (fluent)

EDUCATION

University of California, Berkeley

Master of Engineering in Materials Science & Engineering

08/23 – 05/25

Berkeley, CA

Harvey Mudd College

Bachelor of Science in Computer Science

08/17 – 05/21

Claremont, CA